

Business Intelligence in Global Geopolitics

Spatiotemporal Analysis of Conflicts and Arms Flows

Gabriel Pinto

Joao Fonseca

Jose Cunha

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Abstract

This project develops a Business Intelligence solution to analyse the relationship between international arms transfers and organized political violence. It integrates a SIPRI-based arms transfer dataset with the UCDP Georeferenced Event Dataset, transforms both sources through Power Query, models them using a star schema and presents the results in Power BI dashboards. The final solution supports descriptive analytics, diagnostic analysis and exploratory data mining through country-year aggregation, temporal lag variables, hotspot analysis, heuristic risk scoring and anomaly-oriented decision support.

1 Introduction and Objectives

The project addresses the problem of understanding how international arms flows relate to conflict intensity across countries and time. The objective is not to prove direct causality, but to build a decision-support system capable of combining arms transfer records with geolocated conflict events and revealing relevant analytical patterns.

The main analytical questions are:

- Which countries receive the largest volumes of arms and which countries register the highest fatality levels?
- How do arms deliveries and fatalities evolve over time?
- Are arms delivered in previous years associated with higher conflict intensity in later years?
- Which country-year observations show high-risk or anomalous patterns?

From a BI perspective, the project has three goals: descriptive analytics, diagnostic exploration and exploratory data mining. Descriptive analytics summarize arms flows and conflict fatalities. Diagnostic analytics compare arms deliveries and fatalities over time. Exploratory data mining identifies country-year observations that combine high conflict intensity, frequent events and relevant previous arms deliveries. Since the sources are observational and heterogeneous, the system supports analytical reasoning rather than automatic causal claims.

Requirement	Implementation in the final solution
Descriptive analytics	KPIs, rankings, trends, maps and arms flow analysis.
Diagnostic and exploratory data mining	Country-year aggregation, lag variables, heuristic risk score and anomaly-oriented scatter plots.
Interaction by time and geography	Slicers by year, country, region, weapon category and violence type.
Data Warehouse design	Star schema with conformed dimensions and explicit fact table grains.
ETL automation	Power Query transformations defined inside the PBIX refresh pipeline.
Validation and performance metrics	Row counts, unmatched key checks, source/load metrics and refresh measurement status.
Performance optimization	Reduced loaded columns, disabled raw staging load, star schema, country-year pre-aggregation and explicit DAX measures.
Decision support	Hotspot detection, risk-oriented ranking, anomaly detection and exploratory early-warning analysis.

Table 1: Mapping between project requirements and implementation choices.

Figure 1 summarizes the overall pipeline followed in the project, from the original source files to the analytical dashboards and decision-support layer.

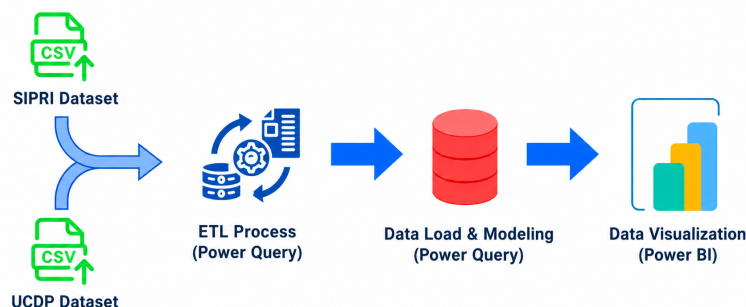


Figure 1: Overall BI pipeline followed in the project, from source CSV files to ETL, dimensional modelling, OLAP dashboards and decision-support analysis.

2 Data Sources

Two heterogeneous datasets were used. The first describes international arms transfers and is based on the [SIPRI Arms Transfers Database](#). It covers the period between 2000 and 2023 and is used to analyse the direction, volume and category of military equipment flows between exporting and importing countries.

The second dataset is the [UCDP Georeferenced Event Dataset](#), produced by the [Uppsala Conflict Data Program](#). It records individual conflict events with geographic and temporal precision, including country, region, date, coordinates, actors, type of violence and estimated fatalities. In this project, the UCDP GED dataset covers 1989–2024.

Dataset	Rows	Columns	Main attributes
Global Arms Transfer	10,520	15	source, target, order date, delivery date, quantity, weapon category, coordinates
UCDP GED Event	385,918	49	country, region, date, coordinates, conflict, dyad, violence type, fatalities

Table 2: Source datasets used in the project.

The sources differ in granularity: one arms record represents a transfer, while one UCDP record represents a conflict event. This motivated the creation of separate fact tables and a country-year analytical table. The arms dataset defines flows between suppliers and recipients through fields such as **source**, **target**, **delivery_date**, **quantity**, **designation** and **category**. The conflict dataset contains event-level information, including conflict identifiers, actors, violence type, coordinates and fatality estimates. The fields **best**, **high** and **low** support central fatality estimates and uncertainty measurement.

The main data quality issues were the absence of a shared country identifier, inconsistent country naming and different temporal coverage. Country names therefore had to be normalized, and combined interpretation is most reliable in the overlapping period between 2000 and 2023.

3 ETL, Refresh and Performance

The ETL process was implemented in Power Query inside Power BI Desktop. Original CSV queries were kept as staging tables, while clean queries were created for transformation and model construction. The staging layer was disabled from final loading to avoid exposing raw data directly in the analytical model.

Transformation	Purpose
Type conversion	Converts dates and numeric fields using explicit type handling.
Text reduction	Removes heavy textual fields such as articles and headlines from the final model.
Country aliases	Normalizes country names and prevents failed joins between datasets.
Date keys	Creates order, delivery, start and end date keys for DimDate .
Quality flags	Identifies records with available delivery dates and coordinates.
Fatality uncertainty	Computes the difference between high and low fatality estimates.
Country-year aggregation	Avoids direct many-to-many joins between events and transfers.
Lag variables	Supports diagnostic and exploratory data mining analysis.

Table 3: Main ETL transformations and rationale.

The main implementation challenge was integrating datasets with different structures and grains. A direct join between transfers and conflict events would duplicate values and produce misleading aggregations. The solution keeps separate atomic fact tables and creates **FactCountryYearRisk** as a common analytical layer. Other challenges included country normalization, temporal mismatch, the unequal meaning of arms quantities across weapon categories and source limitations such as missing illicit flows, small arms, battlefield capture and domestic production.

The refresh strategy is based on replacing the source CSV files with newer versions and refreshing the Power BI model. Since transformations are defined in Power Query, the cleaning, normalization, dimensional generation and fact table generation steps are reapplied automatically. The refresh frequency is manual and dataset-driven, normally when SIPRI or UCDP publish updated files. Dimensions follow a Type 1 update strategy, while fact tables are rebuilt from clean staging queries. A full refresh is acceptable for the current project size; in production, incremental refresh could partition conflict events by year and arms transfers by delivery year.

Metric	Observed value
Arms source dataset size	10,520 rows and 15 columns
Conflict source dataset size	385,918 rows and 49 columns
Loaded arms fact size	10,292 records in FactArmsTransfer
Loaded conflict fact size	385,918 records in FactConflictEvent
Loaded country-year analytical table	approximately 4,000 records in FactCountryYearRisk
First load elapsed time	5 minutes and 17 seconds
Subsequent load size	Not measured; only one official version of each dataset was used
Update elapsed time	Not measured; no subsequent official refresh cycle was executed

Table 4: Source size, first load size and update measurement status.

Future uploads are expected to affect the conflict dataset more than the arms dataset because UCDP GED is much larger. The arms CSV has about 10,520 rows and approximately 2 MB, while the UCDP GED CSV has 385,918 rows and approximately 250 MB. New uploads may append new yearly records or correct historical conflict coding. The design reduces the impact of future growth by

excluding heavy text fields, disabling raw staging load and using the compact **FactCountryYearRisk** table for diagnostic visuals. Power BI Desktop was used for ETL, dimensional modelling and OLAP visualization. Power Query was selected because it is integrated into Power BI and keeps the transformation process reproducible inside the PBIX file. No external database was used; storage was handled by the Power BI in-memory semantic model. Although this is not a physical relational Data Warehouse, the model follows DW principles: dimensions, fact tables, explicit grains, surrogate keys and reusable measures. The design could later be migrated to a relational database, Power BI Dataflow, Fabric Lakehouse or cloud warehouse.

Performance optimization was addressed at the semantic model and ETL levels. Physical SQL indexes, views and database partitioning were not implemented because the final solution was not deployed in a relational database. Instead, the project optimized performance by removing heavy columns, disabling raw staging load, using a star schema, avoiding many-to-many joins, keeping fact tables narrow, using conformed dimensions, creating explicit DAX measures and pre-aggregating diagnostic data into **FactCountryYearRisk**.

4 Data Warehouse and Star Schema

The final model was designed as a dimensional model based on star schema principles. More specifically, it follows a fact constellation structure, where multiple fact tables share common dimensions such as country and date. Instead of using the raw datasets directly, the model separates descriptive context from measurable events. This improves interpretability, avoids unnecessary many-to-many relationships and makes the model easier to query through Power BI visuals and DAX measures.

The project intentionally avoids a single central fact table because arms transfers, conflict events, country-year risk indicators and prediction results represent different analytical processes. An arms transfer is not the same observation as a conflict event, and a prediction result is not an individual real-world event. The solution therefore uses two atomic fact tables, one aggregated analytical fact table and one prediction-result fact table.

The two atomic fact tables preserve the original granularity of the main datasets. **FactArmsTransfer** stores one row per arms transfer, while **FactConflictEvent** stores one row per conflict event. This allows detailed drill-down analysis by country, date, weapon category, conflict, dyad and violence type.

The most important modelling decision is the creation of **FactCountryYearRisk**. This table is not a replacement for the atomic facts; it is an aggregated analytical fact table at country-year granularity. It allows arms deliveries and conflict outcomes to be compared at the same level of detail without multiplying transfer quantities by the number of conflict events in the same country. In other words, the model preserves detail for drill-down while using aggregation only where comparison is analytically valid.

The model also includes **FactPredictionResult**, which stores the outputs of the predictive component at country-year-model granularity. This table allows the comparison between actual deaths and predicted deaths, as well as the calculation of error metrics such as absolute error, squared error and prediction bias.

The **PredictionFeatureImportance** table was intentionally kept as a disconnected auxiliary table. It describes the relative importance of the input variables used by the predictive model and does not share the same analytical grain as the country-year prediction results. Therefore, it is used only for explanatory visualisations and does not need to filter, or be filtered by, the main dimensional model.

Table	Grain	Purpose
DimCountry	one normalized country/entity	Shared geography filter for countries and regions.
DimDate	one day	Supports year, month, quarter, semester and decade analysis.
DimWeapon	one weapon designation/category	Supports analysis by weapon type, category and designation.
DimViolenceType	one violence type	Maps UCDP violence codes to readable analytical labels.
DimConflict	one conflict/dyad combination	Stores conflict, dyad, SideA and SideB .
FactArmsTransfer	one arms transfer	Measures transferred quantity, number of transfers and delivery lag.
FactConflictEvent	one conflict event	Measures deaths, events, civilian deaths and fatality uncertainty.
FactCountryYearRisk	one country-year	Supports lag analysis, risk scoring and exploratory comparison between arms and conflict indicators.
FactPredictionResult	one country-year-model prediction	Stores actual deaths, predicted deaths and prediction error metrics.
PredictionFeatureImportance	one model-feature pair	Static auxiliary table used to explain the importance of predictive input variables.

Table 5: Final dimensional model and table granularities.



Figure 2: Final dimensional schema with primary keys, foreign keys, data types and fact table granularities.

A key modelling decision is the reuse of shared dimensions across different analytical roles. **DimCountry** acts as a role-playing dimension in **FactArmsTransfer**. The same country dimension is used twice: **SourceCountryKey** identifies the exporting country, while **TargetCountryKey** identifies the importing country. The role is defined by the foreign key column in the fact table, while **DimCountry** stores the standardized country name and region.

The same principle is applied to dates. **DimDate** is reused for order dates, delivery dates, conflict start dates, conflict end dates and country-year analytical dates. This avoids creating separate date dimensions and keeps time-based filtering consistent across the model. The main active relationships reflect the default analytical interpretation. **FactArmsTransfer** is actively related to **DimCountry** by target country because the main arms analysis focuses on importing countries. The source-country relationship is kept inactive and used through DAX measures such as **USERRELATIONSHIP** when supplier-specific analysis is required. Similarly, delivery date is the default active relationship for arms transfers, while order date is kept inactive; conflict start date is active for conflict events, while conflict end date is kept inactive.

Relationship	Status	Reason
DimCountry to FactArmsTransfer by target country	Active	Main arms analysis is by importing country.
DimCountry to FactArmsTransfer by source country	Inactive	Used for supplier-specific measures.
DimDate to FactArmsTransfer by delivery date	Active	Delivery is more relevant for comparison with conflict outcomes.
DimDate to FactArmsTransfer by order date	Inactive	Useful for procurement and ordering analysis.
DimCountry to FactConflictEvent	Active	Conflict events are analysed by country.
DimDate to FactConflictEvent by start date	Active	Main conflict timeline uses event start.
DimDate to FactConflictEvent by end date	Inactive	Kept for completeness and duration analysis.
DimCountry to FactCountryYearRisk	Active	Risk indicators are aggregated by country.
DimDate to FactCountryYearRisk by year date	Active	Risk indicators are aggregated by year.
DimCountry to FactPredictionResult	Active	Prediction results are analysed by country.
DimDate to FactPredictionResult by year date	Active	Prediction results are analysed by year.

Table 6: Main active and inactive relationship choices.

The dashboards rely on explicit DAX measures instead of implicit column aggregation. The most important measures are **Total Arms Quantity**, **Total Arms Quantity by Source**, **Total Deaths**, **Total Conflict Events**, **Civilian Death Share**, **Average Delivery Lag**, **Risk Arms Lag 1**, **Risk Arms Lag 2**, **Risk Arms Lag 3**, **Average Risk Score**, **MAE**, **MAPE**, **R2** and **Prediction Bias**. These measures make calculations reusable, easier to validate and consistent across country, year, region, weapon category, violence type and model filters.

This structure also improves maintainability. If new CSV versions add countries, conflicts or weapon designations, the dimensions can

grow without changing the analytical logic. If new dashboard pages are added, they can reuse the same facts, dimensions and measures instead of creating isolated transformations for each visual.

5 Big Questions and Query Logic

The star schema was designed to answer the main DW and data mining questions through combinations of dimensions, fact measures and aggregation levels. Dimensions define the perspective of analysis, while fact tables provide the measurable quantities.

Big Question	Main tables	Logic / expected output
How do arms imports vary over time and geography?	FactArmsTransfer, DimCountry, DimDate	Aggregate arms quantity by target country, region and delivery year.
Where are conflict hotspots concentrated?	FactConflictEvent, DimCountry, DimViolenceType	Aggregate deaths and events by country, region and violence type.
Which weapon categories dominate imports?	FactArmsTransfer, DimWeapon, DimCountry	Group transfer quantities by weapon category and importing country.
Do arms deliveries precede higher conflict intensity?	FactCountryYearRisk, DimCountry	Compare fatalities with ArmsQty_Lag1, Lag2 and Lag3.
Which cases are anomalous or high risk?	FactCountryYearRisk, DimCountry	Rank country-year observations by RiskScore, deaths and arms variables.

Table 7: Mapping between analytical questions and star schema query logic.

The key advantage of this structure is that every question follows a controlled relationship path. Descriptive questions use the atomic fact tables, while diagnostic questions use the country-year aggregate. This separation prevents over-counting and makes the SQL/DAX logic easier to audit. For example, an arms importer ranking should not join directly to individual conflict events, because that would repeat the same transfer for every event in the country. Instead, the model answers transfer questions from FactArmsTransfer, conflict questions from FactConflictEvent, and combined temporal questions from FactCountryYearRisk.

Although the final solution is implemented in Power BI, the same star schema could be loaded into a relational database. For example, the diagnostic country-year logic could be expressed as:

```
SELECT
  c.CountryName,
  r.Year,
  r.TotalDeaths,
  r.TotalEvents,
  r.TotalArmsDelivered,
  r.ArmsQty_Lag1,
  r.ArmsQty_Lag2,
  r.ArmsQty_Lag3,
  r.RiskScore
FROM FactCountryYearRisk r
JOIN DimCountry c
  ON r.CountryKey = c.CountryKey
WHERE r.Year BETWEEN 2000 AND 2024
ORDER BY r.RiskScore DESC;
```

This illustrates that the analytical logic is not dependent on a specific dashboard. The dimensional model supports descriptive queries, diagnostic lag analysis and anomaly detection through simple joins and aggregations.

6 Dashboards and OLAP

The final Power BI report was organized into eight dashboard pages connected by the same dimensional model. Users can interact with the OLAP layer through slicers by year, region, country, weapon category, violence type and prediction model set. The report follows an OLAP-style logic: users slice the data by geography, time, conflict type, weapon category, actor or model output, and then compare the same measures across multiple analytical perspectives. In the current academic implementation, the report is accessed as a PBIX file in Power BI Desktop; in production, it could be published to Power BI Service.

Page	Objective	Main visuals / question answered
Overview	Executive summary	KPIs, yearly comparison, top countries. Identifies global scale and concentration.
Arms Flows	Arms transfer analysis	Suppliers, importers, delivery trends and weapon categories. Separates exporter and importer roles.
Conflict Hotspots	Spatial conflict analysis	Map, regional charts and violence types. Shows where fatalities and events concentrate.
Conflict Actors	Actor-level analysis	Rankings by SideA, SideB and DyadName. Identifies fatal actor relationships.
Temporal Diagnosis	Lag-based diagnosis	Fatalities versus same-year and previous-year arms deliveries. Supports temporal comparison.
Risk / Anomalies	Prioritization	Highest-risk country-years, scatter plots and high deaths without same-year arms.
Prediction	Predictive model evaluation	Random Forest prediction results, error metrics, actual versus predicted fatalities and feature importance.
Validation	Technical validation	Row counts, unmatched keys and consistency checks between detailed and aggregated facts.

Table 8: Power BI dashboard pages and analytical role.

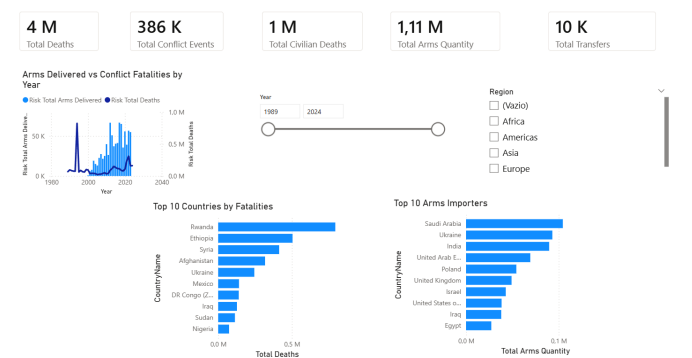


Figure 3: Overview dashboard: global KPIs, yearly comparison and top countries.



Figure 5: Conflict Hotspots dashboard: fatalities by geography, region and violence type.

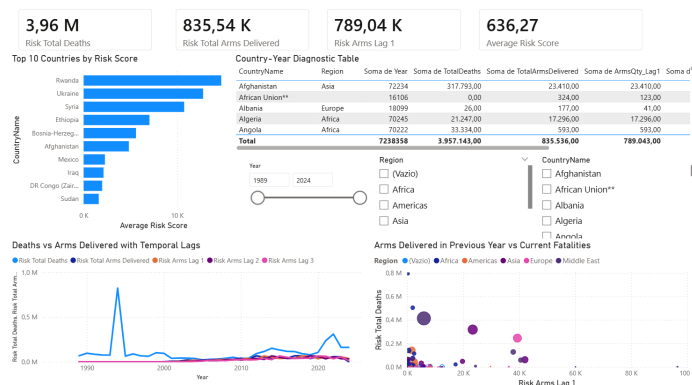


Figure 7: Temporal Diagnosis dashboard: fatalities compared with same-year and lagged arms deliveries.

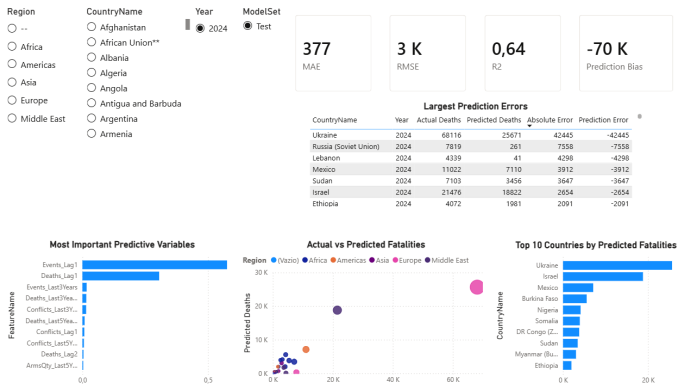


Figure 9: Prediction dashboard: Random Forest evaluation, prediction errors and feature importance.

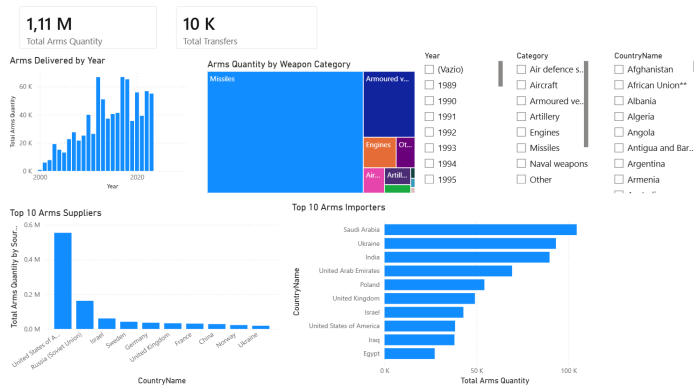


Figure 4: Arms Flows dashboard: suppliers, importers, delivery trends and weapon categories.

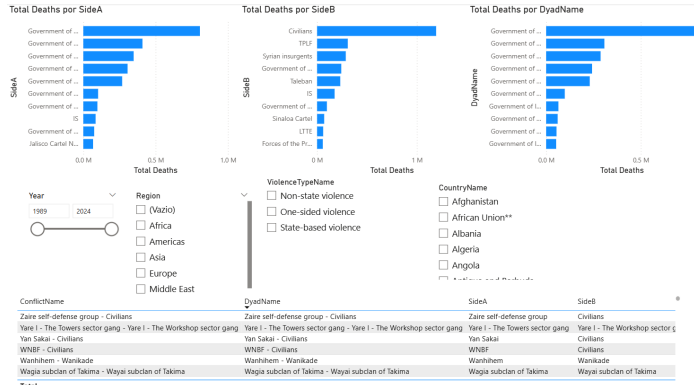


Figure 6: Conflict Actors dashboard: fatality rankings by Side A, Side B and dyad.

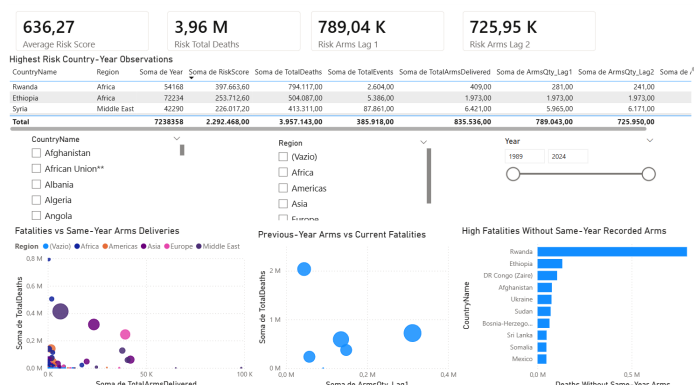


Figure 8: Risk / Anomalies dashboard: highest-risk country-years and anomaly-oriented scatter plots.

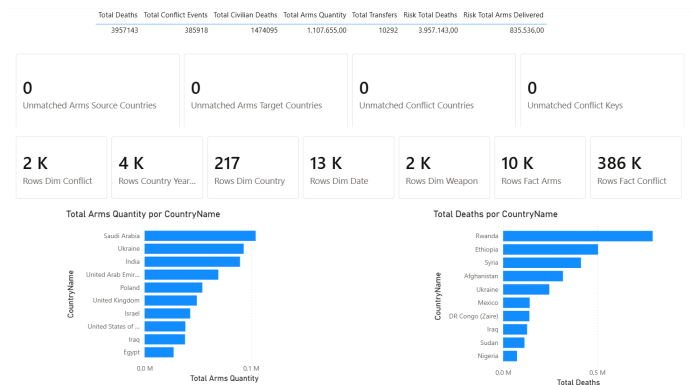


Figure 10: Validation dashboard: row counts, unmatched key checks and consistency checks between facts.

The dashboard pages were designed to move from global overview to detailed investigation. The **Overview**, **Arms Flows** and **Conflict Hotspots** pages support descriptive analytics. The **Temporal Diagnosis** and **Risk / Anomalies** pages support diagnostic analysis and exploratory data mining. The **Conflict Actors** page adds actor-level interpretation. The **Prediction** page extends the analysis with a Random Forest model, allowing the comparison of actual and predicted fatalities, the inspection of prediction errors and the interpretation of the most important predictive variables. Finally, the **Validation** page confirms technical consistency. The **Prediction** dashboard uses the `FactPredictionResult` table to evaluate the Random Forest model on the test set. It includes global performance indicators such as MAE, RMSE, R^2 and prediction bias, together with a table of the largest absolute errors. The scatter plot compares actual and predicted fatalities by country and region, making it possible to identify under-predicted and over-predicted cases. The dashboard also uses the static `PredictionFeatureImportance` table to show which variables contributed most to the model,

with lagged conflict indicators such as previous events and deaths appearing as the most relevant predictors.

7 Diagnostic Analysis and Exploratory Data Mining

The diagnostic component uses **FactCountryYearRisk**, because this table combines arms transfers and conflict events at the same level of granularity: one country in one year. This avoids direct many-to-many comparison between event-level conflict records and transfer-level arms records. It also makes the analysis more efficient, since temporal and risk visuals query a compact country-year table instead of repeatedly aggregating hundreds of thousands of event rows. The data mining component was divided into two complementary parts. The first part was exploratory and anomaly-oriented, based on temporal lag variables and a heuristic risk score. Its objective was not to establish causality, but to discover country-year patterns that may deserve further investigation. The second part was predictive, using a Random Forest regression model to estimate fatalities from historical conflict and arms-related indicators.

Variable group	Examples	Use in the diagnostic layer
Conflict intensity	TotalDeaths , TotalCivilianDeaths	Measures scale, frequency and civilian impact of violence.
Arms exposure	TotalArmsDelivered , ArmsQty_Lag1/2/3	Compares current conflict outcomes with recent and delayed arms deliveries.
Uncertainty	AvgFatalityUncertainty	Identifies observations where fatality estimates are less precise.
Risk ranking	RiskScore	Produces an interpretable shortlist of country-years for deeper investigation.
Prediction features	Events_Lag1 , Deaths_Lag1 , ArmsQty_Lag1/2/3	Used by the Random Forest model to estimate fatalities and evaluate predictive patterns.

Table 9: Main variables used in diagnostic analysis and exploratory data mining.

Three arms lag variables were created:

- **ArmsQty_Lag1**: arms delivered one year before the observed country-year;
- **ArmsQty_Lag2**: arms delivered two years before;
- **ArmsQty_Lag3**: arms delivered three years before.

The exploratory risk score is:

$$\text{RiskScore} = 0.5 \times \text{TotalDeaths} + 0.3 \times \text{ArmsQty_Lag1} + 0.2 \times \text{TotalEvents}$$

TotalDeaths captures conflict intensity, **ArmsQty_Lag1** captures previous-year arms deliveries and **TotalEvents** captures event frequency. The weights were chosen heuristically to prioritize conflict severity while still including previous arms deliveries and event frequency.

This score is not a predictive model and does not establish causality. It is an exploratory data mining technique for anomaly detection and prioritization. It helps identify cases where high fatalities, frequent conflict events and previous arms deliveries appear together. Its main limitation is that the components are not normalized, meaning variables with larger numeric scales, especially deaths, can dominate the score.

The Risk / Anomalies dashboard supports two interpretations. First, country-years with high previous-year arms deliveries and high current fatalities may indicate cases where arms inflows preceded conflict escalation. Second, country-years with high fatalities but low same-year recorded deliveries may indicate accumulated stockpiles, informal routes, missing transfer records, domestic production or conflict dynamics driven by other factors.

The analysis was designed as a triage mechanism. A high-risk observation should not be read as a conclusion by itself; it should be read as a signal that the country-year deserves additional contextual investigation. For example, an analyst can compare the ranked country-year with the arms flow page, inspect the dominant weapon categories, check whether fatalities are concentrated in a specific conflict dyad, and then validate the interpretation with external geopolitical sources.

The most important methodological control is the use of lag variables. Same-year comparison can be misleading because arms can be ordered before delivery, delivered before use, or remain in stockpiles for years. By comparing fatalities with **ArmsQty_Lag1**, **Lag2** and **Lag3**, the report does not assume an immediate relationship and instead supports temporal exploration across several plausible delay windows.

7.1 Predictive Data Mining with Random Forest

In addition to the heuristic risk score, a predictive data mining component was implemented using a Random Forest regression model. The objective of this component was not to prove a causal relationship between arms transfers and conflict fatalities, but to test whether historical country-year indicators contain useful predictive signal for estimating future or held-out fatality values. The prediction task was structured at country-year level, using the same analytical granularity as **FactCountryYearRisk**. This ensured that the predictive model did not directly mix event-level conflict rows with transfer-level arms rows. Instead, each observation represented one country in one year, with aggregated conflict indicators, arms delivery indicators and lagged variables. The model outputs were stored in **FactPredictionResult**. This table contains the actual fatality value, the predicted fatality value and error metrics such as prediction error, absolute error and squared error. These outputs are used in the Prediction dashboard to evaluate model performance and identify cases where the model under-predicts or over-predicts fatalities.

Component	Table / metric	Purpose
Prediction result	FactPredictionResult	Stores actual deaths, predicted deaths and prediction errors by country, year and model set.
Feature importance	PredictionFeatureImportance	Stores the relative importance of each input variable used by the Random Forest model.
Model evaluation	MAE, RMSE, R^2 , Prediction Bias	Evaluates average error, large-error sensitivity, explained variance and systematic under/over-prediction.
Error analysis	Largest prediction errors	Identifies countries and years where the model performs worst and where contextual investigation is needed.

Table 10: Predictive data mining components used in the Random Forest analysis.

The Prediction dashboard reports four main performance indicators. **MAE** measures the average absolute prediction error and is easier to interpret in the original fatality scale. **RMSE** penalizes large errors more strongly and is useful for detecting major prediction failures. **R^2** summarizes the share of variance explained by the model. **Prediction Bias** indicates whether the model tends to over-predict or under-predict fatalities overall.

The feature importance visual is based on the static **PredictionFeatureImportance** table. This table is intentionally disconnected from the main dimensional model because it describes the internal behaviour of the Random Forest model rather than a country-year event. In the current results, lagged conflict variables such as previous events and previous deaths appear as the strongest predictors, suggesting that recent conflict history is more informative than arms indicators alone.

The largest prediction errors are analytically important. For example, if a country has very high actual deaths but much lower predicted deaths, this indicates that the model failed to anticipate the scale of violence from the available historical features. These cases should not be treated only as model errors; they can also be used as anomaly signals for deeper qualitative analysis.

The predictive component has several limitations. First, the model is trained on observational data and therefore cannot establish causality. Second, conflict fatalities are highly skewed, with a small number of extreme country-years accounting for a large share of deaths. Third, many geopolitical factors that influence conflict escalation are not captured in the dataset, such as alliances, military strategy, domestic politics, informal arms routes or external interventions. Finally, the model results depend on the selected features, the train/test split and the temporal structure of the data.

Overall, the data mining layer combines interpretability and prediction. The heuristic risk score provides a transparent ranking of country-years for triage and investigation, while the Random Forest model provides a predictive benchmark and highlights which historical variables are most useful for estimating fatalities. Both components are exploratory and diagnostic rather than causal.

8 Persona and Decision Support

To demonstrate decision support, the workflow of a conflict risk analyst was simulated. The analyst works in an international security monitoring unit and wants to identify countries where recent arms flows, conflict hotspots and actor patterns suggest potential escalation or require further investigation.

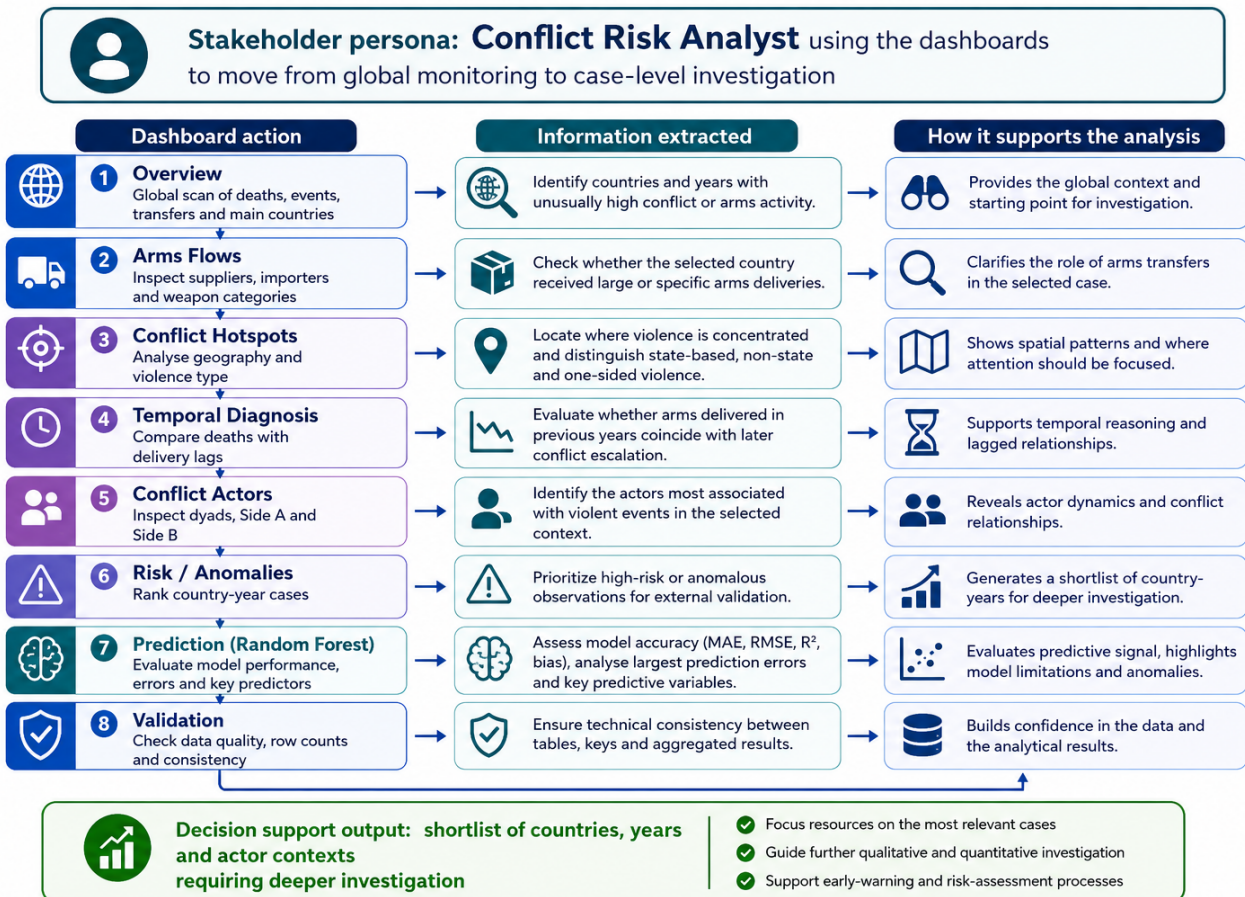


Figure 11: Persona-based dashboard usage simulation from global monitoring to case-level decision support.

The analyst starts with the Overview dashboard to identify global trends and high-fatality or high-import countries. She then uses Arms Flows to inspect suppliers, recipients and weapon categories for a selected country. Conflict Hotspots provides spatial and

violence-type context, while Temporal Diagnosis compares fatalities with arms deliveries from previous years. Conflict Actors identifies relevant dyads and actor sides. Finally, Risk / Anomalies ranks country-year observations that should be investigated using external geopolitical knowledge.

The final model supports decision-making in three ways. First, it identifies global hotspots where conflict fatalities are concentrated. Second, it describes the structure of arms flows, including suppliers, importers and weapon categories. Third, it enables diagnostic and exploratory data mining by comparing conflict intensity with same-year and previous-year arms deliveries. The system is therefore an exploratory early-warning and prioritization tool, not an automated decision system.

9 Findings, Limitations and Future Work

The final BI solution showed that arms transfers and conflict fatalities are both highly concentrated, but not necessarily concentrated in the same countries. A small number of countries account for a large share of recorded arms imports, while another group accounts for the highest fatality totals. Top arms importers include Saudi Arabia, Ukraine, India, United Arab Emirates and Poland, whereas top countries by fatalities include Rwanda, Ethiopia, Syria, Afghanistan and Ukraine. This confirms that arms-flow intensity and conflict severity must first be analysed separately before being compared in a combined country-year layer.

A second important finding is that Ukraine appears as a relevant case in both domains. It is among the main arms importers and also among the countries with high conflict fatalities. This does not prove a causal relationship, but it shows the value of the integrated BI model: the analyst can move from a global ranking to a specific country, inspect its arms inflows, compare them with conflict outcomes, and then examine the temporal and actor-level context.

The model also showed that many high-fatality countries are not necessarily the largest arms importers in the official transfer dataset. This is analytically important because it suggests that conflict intensity cannot be explained only by recorded international arms transfers. Other mechanisms may be involved, including historical stockpiles, domestic production, illicit flows, captured weapons, regional spillovers, political instability or conflict dynamics that are not represented in the arms transfer data.

The Arms Flows dashboard revealed that arms transfers are structurally asymmetric. A limited number of supplier and recipient countries dominate the network, and the interpretation of arms quantity depends strongly on the weapon category. The same quantity value can represent very different military capabilities, since one aircraft, one missile system and one vehicle are all counted as units. Therefore, total arms quantity is useful for scale detection, but category-level analysis is necessary for meaningful interpretation.

The Conflict Hotspots and Conflict Actors dashboards showed that conflict fatalities are geographically and actor-wise concentrated. Fatalities are not evenly distributed across countries, regions or conflict dyads. The inclusion of `DimConflict`, `SideA`, `SideB` and `DyadName` was therefore important because it allowed the analysis to move beyond country totals and identify which actor relationships contributed most to violent outcomes.

The Temporal Diagnosis dashboard showed that same-year comparison is insufficient. Arms transfers can be ordered before delivery, delivered before use, stored for several years, or used indirectly through existing stockpiles. For this reason, the use of lag variables was necessary. The comparison between fatalities and `ArmsQty_Lag1`, `ArmsQty_Lag2` and `ArmsQty_Lag3` provided a more realistic diagnostic view than a simple same-year comparison.

The Risk / Anomalies dashboard produced a practical decision-support output: a shortlist of country-years requiring deeper investigation. High-risk cases should not be interpreted as proof of causality, but as analytical signals. They help identify where high fatalities, frequent conflict events and previous arms deliveries appear together. These cases can then be investigated through the other dashboards by checking arms categories, suppliers, conflict dyads, geographic concentration and temporal patterns.

The predictive component added a second data mining perspective. The Random Forest model achieved a moderate predictive performance, with an R^2 of 0.64, a MAE of 377 and an RMSE of around 3,000 on the test set. This indicates that the selected country-year variables contain useful predictive signal, but also that the model struggles with extreme high-fatality cases. The difference between MAE and RMSE suggests that most predictions are reasonably close, while a small number of severe conflicts generate very large errors.

The largest prediction errors are themselves analytically useful. Cases such as Ukraine in 2024 show that the model substantially under-predicted fatalities when violence reached a level that was not well captured by previous historical indicators. These errors should not only be treated as model failures; they can also be interpreted as anomaly signals, highlighting cases where additional geopolitical context is required.

The feature importance results showed that lagged conflict variables, especially previous events and previous deaths, were the strongest predictors in the Random Forest model. This suggests that recent conflict history is more informative for predicting fatalities than arms transfer variables alone. Arms-related variables remain useful as contextual indicators, but the model indicates that conflict persistence and recent violence intensity are the dominant predictive signals.

A further implementation finding is that the predictive layer must be interpreted as a refresh-time analytical output rather than as an interactive re-training system. The Random Forest model is executed in Power Query during dataset refresh, and the dashboard filters only slice the resulting prediction table. Therefore, changing a country, region or year slicer does not re-train the model; it only filters the already generated outputs in `FactPredictionResult`. This design is appropriate for a BI report because it keeps the predictive results stable and auditable, but it also means that model governance, Python environment configuration and refresh reproducibility become part of the analytical architecture.

Overall, the project demonstrates that the main value of the BI system is not to claim that arms transfers cause conflict, but to support structured investigation. The dashboards allow analysts to move from global monitoring to country-level drill-down, from country totals to actor-level patterns, from descriptive analysis to temporal diagnosis, and from heuristic risk scoring to predictive error analysis. The system therefore works as a diagnostic and decision-support tool rather than as a causal inference framework.

The main limitations are:

- different temporal coverage: conflicts cover 1989–2024, while arms transfers cover 2000–2023, meaning that the 2024 predictive test year relies mainly on previous-year arms indicators rather than complete same-year arms-transfer information;
- official arms transfers exclude illicit flows, small arms, captured weapons, domestic production and indirect routes;
- fatality estimates include uncertainty and depend on source quality;

- country names and non-standard entities require normalization;
- arms quantity is not equivalent across weapon categories;
- lag analysis, heuristic risk scoring and predictive modelling are diagnostic and exploratory, not causal;
- the predictive model is based on aggregated observational data and performs less accurately in extreme high-fatality cases;
- the Random Forest model is executed during Power Query refresh, so slicers filter stored prediction outputs rather than dynamically re-training the model;
- reproducible deployment of the Python-based predictive step depends on a controlled Python environment, installed package versions, Power BI privacy settings and refresh configuration;
- important geopolitical variables, such as alliances, regime changes, military strategy, sanctions, peace agreements and foreign interventions, are not included in the current model.

These issues were identified but not fully eliminated. The BI model mitigates them through country normalization, separate fact tables, validation checks, country-year aggregation, lag variables and explicit uncertainty measures. However, it cannot fully solve limitations that originate in the source data itself.

Future work could improve the solution in four directions. First, the risk score could be normalized so that deaths, arms quantities and event counts contribute on comparable scales. Second, additional explanatory variables could be added, including socioeconomic indicators, political stability measures, alliance data, refugee flows, sanctions and military expenditure. Third, the predictive component could be extended with temporal validation, model comparison and alternative algorithms such as regression, gradient boosting, clustering or panel-data methods. Finally, the technical architecture could be migrated from a PBIX-only implementation to a relational database, Power BI Dataflow, Fabric Lakehouse or cloud warehouse, with incremental refresh partitioned by conflict year and arms delivery year.

10 Conclusion

This project evolved from a Power BI prototype into a structured BI solution with an ETL layer, normalized dimensions, explicit fact table grains, validation metrics, diagnostic analytics and exploratory data mining. Its main contribution is the creation of a coherent analytical structure for two heterogeneous geopolitical datasets.

Instead of merging arms transfers and conflict events directly, the solution keeps separate fact tables and uses a country-year analytical table to compare them at a compatible level of granularity. This improves interpretability and reduces the risk of duplicated or misleading measures. Although the model does not prove causality, it helps users identify relevant countries, years, regions, actors and weapon categories for further investigation. The exploratory data mining component, based on lag variables and a heuristic risk score, adds an additional decision-support layer by identifying country-year observations that combine high conflict intensity, frequent events and relevant previous arms deliveries.

The final solution is also technically defensible because it separates raw data, clean staging logic, dimensions, fact tables, validation measures and dashboards. This makes the project reproducible: when the source CSV files are updated, the same Power Query pipeline can rebuild the analytical model and the same measures can be reused across the report. The validation page confirms that the main dimensional relationships are complete and that the analytical model is not simply a set of independent visuals over raw tables.

From a decision-support perspective, the report should be interpreted as an exploratory monitoring tool. It reduces the time needed to identify relevant cases, but final conclusions still require geopolitical context and external evidence. This is appropriate for the project domain because arms transfers, conflict escalation and actor behaviour are complex phenomena that cannot be explained by one dataset or one metric.

With future data updates and additional indicators, the same structure can support broader geopolitical monitoring, more advanced data mining and possible predictive extensions. The current contribution is therefore not only the dashboards themselves, but the reusable BI architecture that makes those dashboards reliable, extensible and analytically transparent.

AI Assistance Disclosure

Generative AI tools were used as support during the development of this project, mainly for planning the report structure, improving written explanations, drafting possible Power Query and DAX logic, and reviewing the clarity of the technical documentation. The AI assistance was used as an auxiliary tool and not as a substitute for the project work. All dataset preparation, Power BI implementation, ETL decisions, star schema design, dashboard construction, validation checks and final interpretations were reviewed, adapted and validated by the project group. The final responsibility for the analysis, technical decisions and conclusions belongs to the authors.